Documentation: Simple Linear Regression Analysis of Gilgit Climate Records

1. Executive Summary

This project conducts a quantitative climate analysis for the city of Gilgit using historical meteorological data. The primary objective is to evaluate the statistical relationship between monthly extreme temperature parameters: specifically, predicting the **Highest Maximum Temperature in Celsius (y)** based on the **Lowest Minimum Temperature in Celsius (x)**.

Using Python's data science ecosystem (pandas, scikit-learn, matplotlib, and seaborn), a Simple Linear Regression (SLR) model was trained on monthly climate aggregates. The model achieved an R^2 score of **0.8996** and a Root Mean Squared Error (RMSE) of **2.9144 °C**, demonstrating a strong positive linear correlation between monthly extreme minimums and maximums in Gilgit.

2. Dataset Overview and Preprocessing

2.1 Dataset Description

The underlying dataset, titled Climate records of Gilgit.csv, contains long-term historical records of temperature and precipitation for Gilgit.

2.2 Data Cleaning and Preprocessing Workflow

The notebook executes the following data preparation steps:

1. **Importing Libraries**: Essential packages for data manipulation (pandas, numpy), visualization (matplotlib.pyplot, seaborn), and machine learning (sklearn.linear_model.LinearRegression, sklearn.metrics) are loaded.
2. **Loading Dataset**: The raw CSV file is read into a pandas DataFrame (df).
3. **Column Standardization**: Leading and trailing whitespaces in column names are stripped using df.columns.str.strip() to ensure clean key indexing.
4. **Filtering Non-Monthly Aggregates**: The dataset contains an 'Annual' summary row alongside individual monthly entries. To avoid biasing the regression model with annual summary data, rows where df['Month'] == 'Annual' are filtered out using df[df['Month'] != 'Annual'].copy().

3. Exploratory Data Analysis & Feature Selection

3.1 Variables Selected

- **Independent Variable (X)**: Lowest Minimum Temperature in Celsius
- **Dependent Variable (y)**: Highest Maximum Temperature in Celsius

3.2 Domain Context

Gilgit experiences significant seasonal variations in temperature typical of high-altitude valley climates in northern Pakistan. Predicting monthly maximum temperatures based on minimum thresholds provides insights into seasonal temperature ranges and extreme weather dynamics.

4. Methodology & Model Implementation

4.1 Statistical Model: Simple Linear Regression

The modeling approach utilizes Ordinary Least Squares (OLS) regression implemented via scikit-learn's LinearRegression class.

The mathematical formulation of the simple linear model is:

$$y = \beta_1 x + \beta_0$$

Where:

- y = Highest Maximum Temperature (°C)
- x = Lowest Minimum Temperature (°C)
- β_1 = Slope coefficient (rate of change in max temperature per unit change in min temperature)
- β_0 = y-intercept (predicted maximum temperature when minimum temperature is 0 °C)

4.2 Training Procedure

- Feature vector X and target vector y are extracted from the filtered monthly DataFrame df_monthly.
- model.fit(X, y) fits the regression plane to the monthly records.
- Predictions are computed via y_pred = model.predict(X).

5. Statistical Results and Model Evaluation

5.1 Key Performance Metrics

Metric	Formula / Variable	Derived Value
Slope (β_1)	model.coef_[0]	**1.2136**
Intercept (β_0)	model.intercept_	**35.6320**

Coefficient of Determination (R^2)	$1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y})^2}$	**0.8996**
Root Mean Squared Error (RMSE)	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	**2.9144 °C**

5.2 Fitted Regression Equation

$$PredictedMaxTemp(°C) = 1.2136 \times (LowestMinTemp°C) + 35.6320$$

5.3 Interpretation

- **Slope ($\beta_1 = 1.2136$)**: For every 1 °C increase in a month's lowest minimum temperature, the predicted highest maximum temperature increases by approximately **1.21 °C**.
- **Intercept ($\beta_0 = 35.6320$)**: During a month where the lowest recorded temperature is 0 °C, the baseline highest maximum temperature is predicted to be **35.63 °C**.
- **R-squared ($R^2 = 0.8996$)**: Approximately **89.96%** of the variance in monthly highest maximum temperatures can be explained by the variation in monthly lowest minimum temperatures.
- **RMSE ($2.9144°C$)**: On average, the model's predictions deviate from actual historical maximums by less than **3 °C**.

6. Visualization & Data Analysis

The project includes a customized visualization script using matplotlib and seaborn:

- **Scatter Plot**: Actual monthly maximum/minimum observations are plotted with dark blue markers (color='darkblue', $s = 80$).
- **Regression Line**: The fitted SLR line is displayed in red (linewidth=2) overlaying the scatter data.
- **Point Annotations**: Each scatter point is annotated with its corresponding month name (plt.annotate) to observe seasonal trends and outliers.
- **Formatting**: Clean titles, axis labels in degrees Celsius, a legend, and custom grid lines (alpha=0.6, linestyle='--') are rendered for clarity.

7. Complete Project Source Code

Below is the complete Python code utilized in the project:

Python
```
import pandas as pd
import numpy as np
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Load Dataset
file_path = "/kaggle/input/datasets/farhankarim11/climate-records-of-gilgit/Climate records of
Gilgit.csv"
df = pd.read_csv(file_path)

# Data Cleaning
df.columns = df.columns.str.strip()
df_monthly = df[df['Month'] != 'Annual'].copy()

# Feature and Target Selection
X = df_monthly[['Lowest Minimum Temperature in Celsius']]
y = df_monthly['Highest Maximum Temperature in Celsius']

# Model Fitting
model = LinearRegression()
model.fit(X, y)

# Predictions & Metrics Calculation
y_pred = model.predict(X)
slope = model.coef_[0]
intercept = model.intercept_
r2 = r2_score(y, y_pred)
rmse = np.sqrt(mean_squared_error(y, y_pred))

# Display Metrics
print("=" * 40)
print("  SIMPLE LINEAR REGRESSION RESULTS")
print("=" * 40)
print(f"Regression Equation : y = {slope:.4f} * x + {intercept:.4f}")
print(f"R-squared (R²)      : {r2:.4f}")
print(f"RMSE                : {rmse:.4f}")
print("=" * 40)

# Visualization
plt.figure(figsize=(10, 6))
sns.scatterplot(
    x='Lowest Minimum Temperature in Celsius',
    y='Highest Maximum Temperature in Celsius',
    data=df_monthly,
    color='darkblue',
    s=80,
    label='Actual Monthly Highs/Lows'
)
```

```
plt.plot(
    df_monthly['Lowest Minimum Temperature in Celsius'],
    y_pred,
    color='red',
    linewidth=2,
    label='SLR Regression Line'
)

# Annotate points with Month names
for i, txt in enumerate(df_monthly['Month']):
    plt.annotate(txt, (X.iloc[i, 0] + 0.5, y.iloc[i] - 0.3), fontsize=8, alpha=0.8)

plt.title('Gilgit Climate: Max vs Min Temperature Linear Regression', fontsize=14, pad=12)
plt.xlabel('Lowest Minimum Temperature (°C)', fontsize=12)
plt.ylabel('Highest Maximum Temperature (°C)', fontsize=12)
plt.legend(loc='upper left')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```[cite: 1]

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8. Key Findings and Future Recommendations

Key Findings
1. **Strong Predictability**: Extreme low temperatures serve as an accurate linear predictor for maximum high temperatures in Gilgit ($R^2 \approx 0.90$)[cite: 1].
2. **Seasonal Variance**: Visual inspection shows predictable clustering by season, with winter months (December, January) residing near lower temperatures and summer months (July, August) at higher extremes[cite: 1].

Future Enhancements
* **Multivariate Modeling**: Incorporate relative humidity, precipitation, and wind speed into a Multiple Linear Regression (MLR) model to lower the RMSE further.
* **Train/Test Splitting**: Incorporate k-fold cross-validation or a train/test split across multi-y